HDOC- 13
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Program to Implement a Basic Calculator Using switch-case

This document provides a clean, well-commented implementation of a basic calculator in C. The program takes two integers and an operator as input and uses a switch-case statement to perform addition, subtraction, multiplication, division or modulus.

1. C Implementation

Step 1: Understand the Logic

- The operator is a single character, so the switch statement compares it with the character cases '+', '-', '*', '/' and '%'.
- Each case performs its operation on the two numbers and ends with break so the next case does not run.
- Division and modulus are not possible when the second number is 0, so these two cases check for zero first and print an error message.
- Division is done as (float)a / b so that the answer keeps its decimal part (for example 10 / 4 = 2.50). Modulus (%) gives the remainder and works only with integers.
- The default case runs when the operator does not match any case, so it reports an invalid operator.

Step 2: Define Variables and Data Types

- Use int variables (a, b) for the two numbers and a char variable (op) for the operator.

Step 3: Take User Input

- Display a prompt using printf("Enter an expression (example: 12 + 5): "), then read the whole expression using scanf("%d %c %d", &a, &op, &b).

Step 4: Apply the switch-case Statement

1. Write switch (op) so that the operator is compared with each case.
2. case '+', '-' and '*': print the sum, difference and product.
3. case '/' and '%': if b == 0 print an error, otherwise print the quotient or the remainder.
4. default: print an error message for any other operator.

Step 5: Format the Output

- Print the complete expression with its result using printf(). A percent sign is printed in printf() by writing %%.

Complete C Code

```c
/*
 * Q20 (User Inputs, Operations & Output)
 * Write a program to implement a basic calculator using
 * switch-case for +, -, *, / and %.
 */
#include <stdio.h>

int main() {
    int a, b;
    char op;

    // Step 1: Read the expression (operand operator operand)
    printf("Enter an expression (example: 12 + 5): ");
    scanf("%d %c %d", &a, &op, &b);

    // Step 2: Perform the operation using switch-case
    switch (op) {
        case '+':
            printf("%d + %d = %d\n", a, b, a + b);
            break;
        case '-':
            printf("%d - %d = %d\n", a, b, a - b);
            break;
        case '*':
            printf("%d * %d = %d\n", a, b, a * b);
            break;
        case '/':
            if (b == 0) {
                printf("Error! Division by zero is not allowed.\n");
            } else {
                printf("%d / %d = %.2f\n", a, b, (float)a / b);
            }
            break;
        case '%':
            if (b == 0) {
                printf("Error! Modulus by zero is not allowed.\n");
            } else {
                printf("%d %% %d = %d\n", a, b, a % b);
            }
            break;
        default:
            printf("Invalid operator! Use +, -, *, / or %%.\n");
    }

    return 0;
}
```

Program Output (Sample Runs)

Run 1:

```
$ gcc hdoc13.c -o hdoc13
$ ./hdoc13
Enter an expression (example: 12 + 5): 12 + 5
12 + 5 = 17
```

Run 2:

```
$ ./hdoc13
Enter an expression (example: 12 + 5): 12 - 5
12 - 5 = 7
```

Run 3:

```
$ ./hdoc13
Enter an expression (example: 12 + 5): 6 * 7
6 * 7 = 42
```

Run 4:

```
$ ./hdoc13
Enter an expression (example: 12 + 5): 10 / 4
10 / 4 = 2.50
```

Run 5:

```
$ ./hdoc13
Enter an expression (example: 12 + 5): 17 % 5
17 % 5 = 2
```

Run 6:

```
$ ./hdoc13
Enter an expression (example: 12 + 5): 8 / 0
Error! Division by zero is not allowed.
```

Run 7:

```
$ ./hdoc13
Enter an expression (example: 12 + 5): 4 ^ 2
Invalid operator! Use +, -, *, / or %.
```

Dry Run with Test Cases

Sample Test Case 1:

- Input: 12 + 5
- Calculation:
  - scanf stores a = 12, op = '+', b = 5
  - switch (op) matches case '+', so the sum is printed: 12 + 5 = 17
  - break exits the switch
- Output: 12 + 5 = 17

Sample Test Case 2:

- Input: 10 / 4
- Calculation:
  - o scanf stores a = 10, op = '/', b = 4
  - o switch (op) matches case '/'; b == 0 is FALSE, so the division is done
  - o (float)10 / 4 = 2.5, which is printed with 2 decimals as 2.50
- Output: 10 / 4 = 2.50

Sample Test Case 3:

- Input: 17 % 5
- Calculation:
  - o scanf stores a = 17, op = '%', b = 5
  - o switch (op) matches case '%'; b == 0 is FALSE, so the modulus is done
  - o 17 % 5 = 2, because 17 = 3 x 5 + 2
- Output: 17 % 5 = 2

Sample Test Case 4:

- Input: 8 / 0
- Calculation:
  - o scanf stores a = 8, op = '/', b = 0
  - o switch (op) matches case '/'; b == 0 is TRUE, so the division is not done
  - o The error message is printed instead
- Output: Error! Division by zero is not allowed.

Sample Test Case 5:

- Input: 4 ^ 2
- Calculation:
  - o scanf stores a = 4, op = '^', b = 2
  - o None of the cases '+', '-', '*', '/' or '%' match '^'
  - o The default case runs and prints the error message
- Output: Invalid operator! Use +, -, *, / or %.